

Confidential

Change Log

Mortgage Hybrid IRB Models – Implementation

Santander Technology | May 2026

Document	Mortgage Hybrid IRB Models – Change Log
Version	1.0
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Introduction

This document records changes and identified discrepancies arising during the **Mortgage Hybrid IRB Models** implementation into the Santander UK Data Lake / Athena platform. It covers two items identified during the reconciliation work of the Working Group (May 2026):

- **Change 1 – sugg_sale_price Condition for PIP Contracts (Not Implemented):** An identified divergence in the LGD code.

1. LGD on Properties in Possession

X NOT IMPLEMENTED – Documented for audit purposes. Requires a future change request.

What happened

During the December 2025 LGD reconciliation, approximately 80% of default accounts had LGDs reconciling within 0.1%. However, a subset of Properties in Possession (PIP) contracts were returning an implemented LGD of 100%. The goal was to modify the initial condition of the LGD recovery valuation logic to treat a zero value in the sugg_sale_price field the same way as a null (missing) value.

In the original implementation, the code treated **0** as a valid recovery value for the Suggested Sale Price. The update requires that if sugg_sale_price is **0**, it should be ignored—just like a null value

The original code is:

```
if default = 1 and pip_ind = 'Y' and sugg_sale_price ne . then do;
    lgd_recovery_valuation = sugg_sale_price;
    lgd_avm_hpi_flag = 'AVM';
end;
else if current_avm_confidence in (., 0) or current_valuation_amt_avm in (., 0)
then do;
    lgd_recovery_valuation = current_valuation_amt;
    lgd_avm_hpi_flag = 'HPI';
end;
else do;
    lgd_recovery_valuation = current_valuation_amt_avm;
    lgd_avm_hpi_flag = 'AVM';
end;
```

Suggest Change:

```
if default = 1 and pip_ind = 'Y' and sugg_sale_price ne . and sugg_sale_price >
0 then do;
    lgd_recovery_valuation = sugg_sale_price;
    lgd_avm_hpi_flag = 'AVM';
end;
else if current_avm_confidence in (., 0) or current_valuation_amt_avm in (., 0)
then do;
    lgd_recovery_valuation = current_valuation_amt;
    lgd_avm_hpi_flag = 'HPI';
end;
else do;
    lgd_recovery_valuation = current_valuation_amt_avm;
    lgd_avm_hpi_flag = 'AVM';
end;
```

Key emails

Date	From	What happened
07 May 2026	Black, Barnaby	Flags that ~131 PIP contracts have LGD = 100% in the system vs ~7% expected. Asks for an explanation.
08 May 2026	Perez Paredes, Jorge	Root cause identified: the system treats a zero sale price as valid, setting the recovery value to £0 and driving LGD to 100%.
11 May 2026	Black, Barnaby	Confirms the issue. Proposes the fix. Quantifies the impact: default LGD rises from 12.76% to 13.84%, capital impact ~£1.5m.
13 May 2026	Perez Paredes, Jorge	Confirms the fix is technically ready but flags it must go through a formal process given the capital impact. Requests a scoping meeting. No decision to proceed received. Change remains on hold. Not accepted on a call after this email.

Thread of the emails:

https://santandernet.sharepoint.com/sites/SANUK-NFQteam/Shared%20Documents/General/08.%20Uk%20Hybrid%20Model/00_Documentation/Changes/RE%20Mortgage%20Hybrid%20Implementation%20-%20Working%20Group.msg

End of document – Mortgage Hybrid IRB Models Change Log v1.0